UCS617

MICROPROCESSOR – BASED SYSTEM DESIGN

LAB ASSIGNMENT – 1 (8086 MICROPROCESSOR)

SUBMITTED TO:

DR. ROHAN SHARMA

SUBMITTED BY:

AKASH PACHAURI (102303105)

FARHAN KAZI (102303994)

JOBAN KHUSHNOOR (102303373)

BHAWESH WAR (102313065)

AGAMJOT SINGH (102303387)

SUBGROUP:

3C31

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

INDEX

Q1.

a) Write an assembly language program to add two 16-bit numbers in 8086 (Fixed Value)

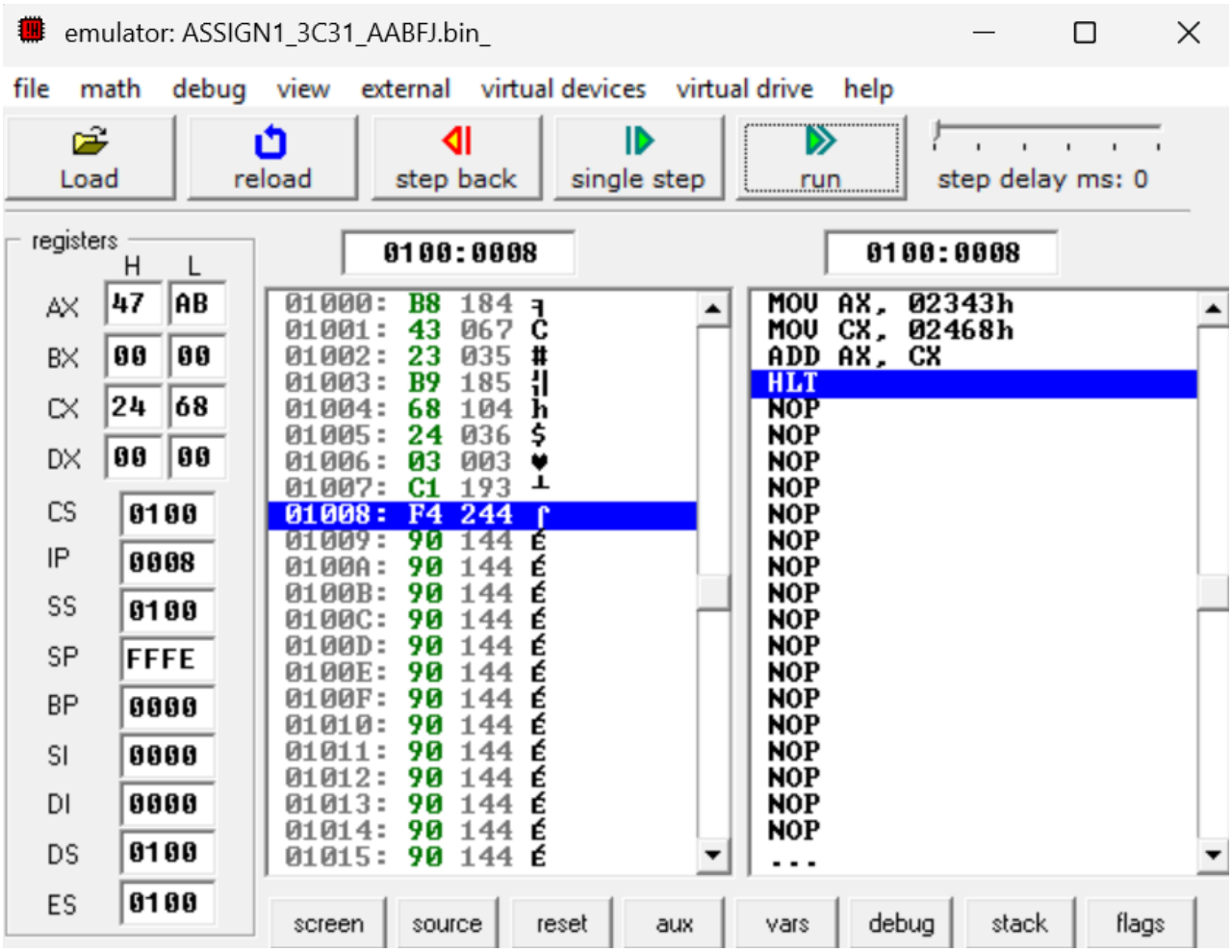
Code:

MOV AX, 2343H

MOV CX, 2468H

ADD AX, CX

HLT



b) Add two 16 - bits hexadecimal numbers (runtime values)

Code:

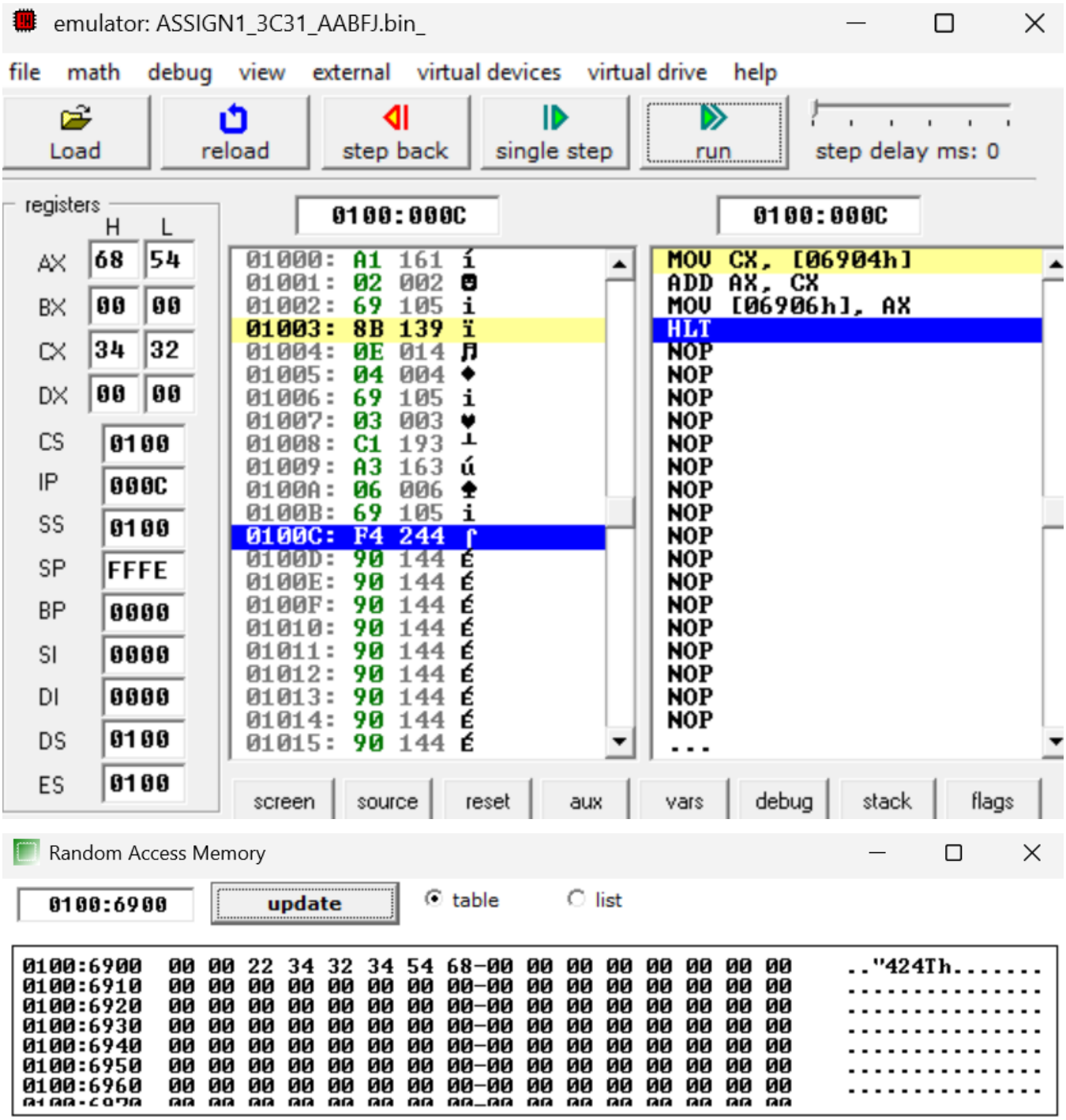
MOV AX, [6902H]

MOV CX, [6904H]

ADD AX,CX

MOV [6906H], AX

HLT

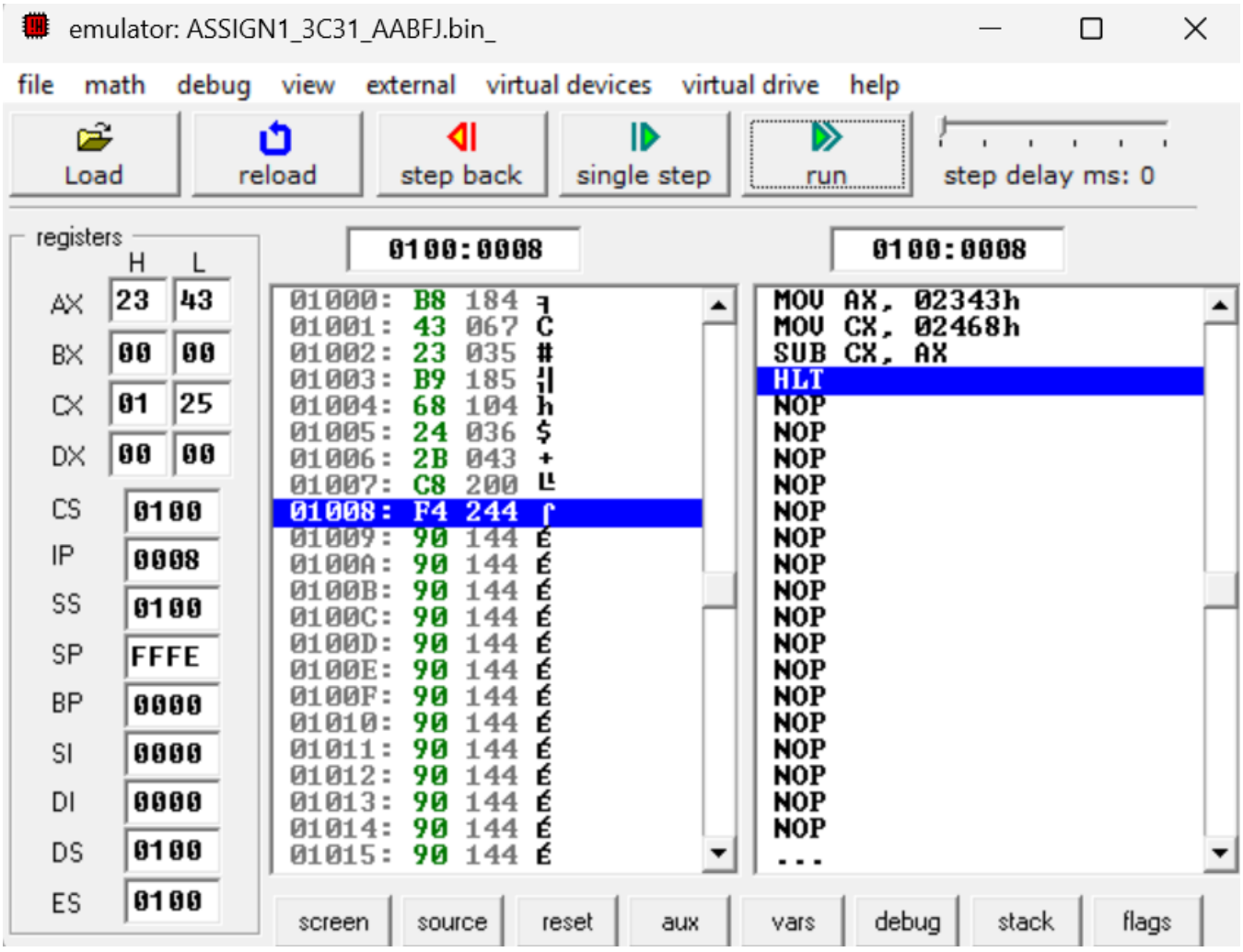


Q2.

a) Write an assembly language program to subtract two 16-bit numbers in 8086. (fixed values)

Code:

```
MOV AX, 2343H
MOV CX, 2468H
SUB CX, AX
HLT
```



b) Subtract two 16 - bits hexadecimal numbers (runtime values)

Code:

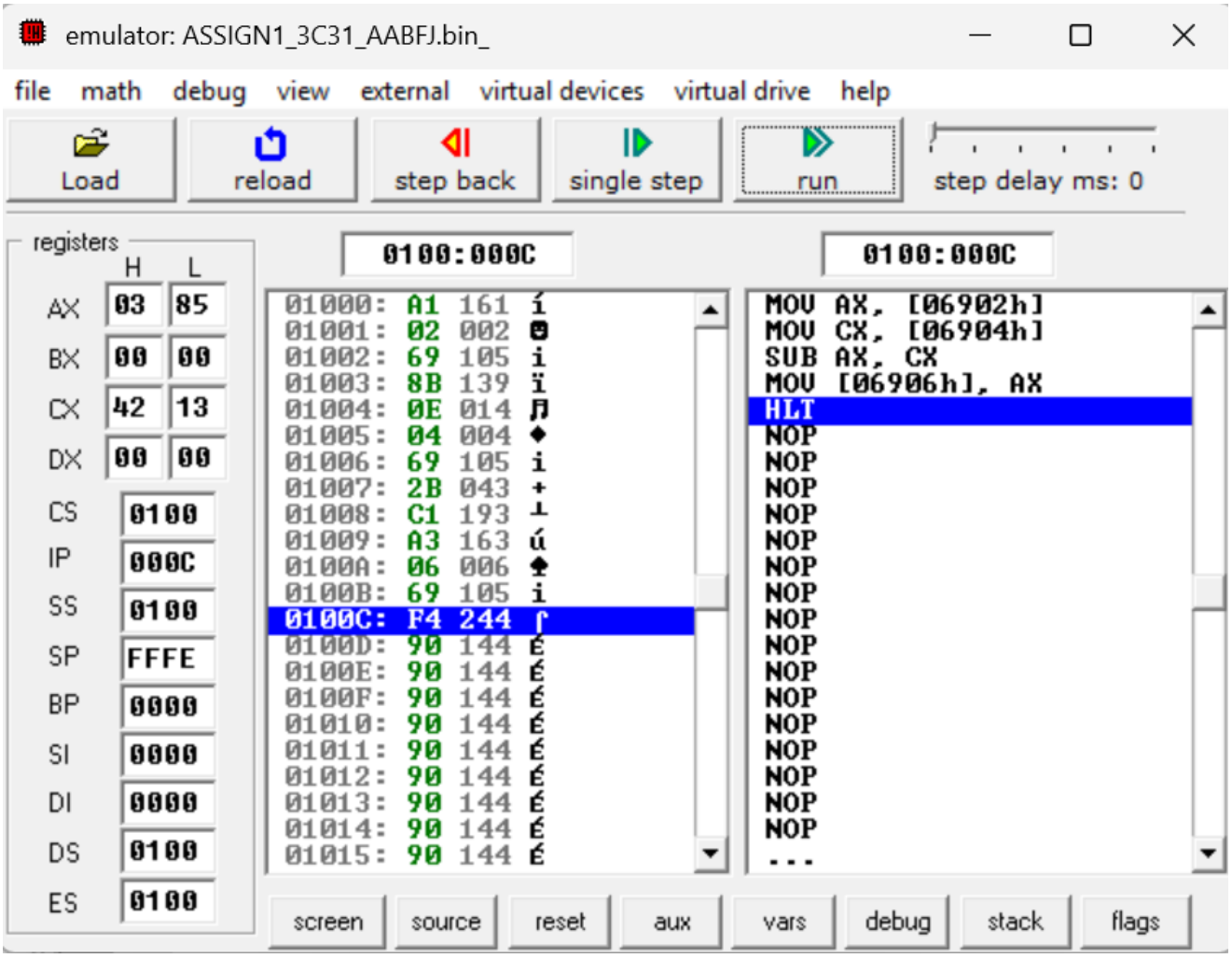
MOV AX, [6902H]

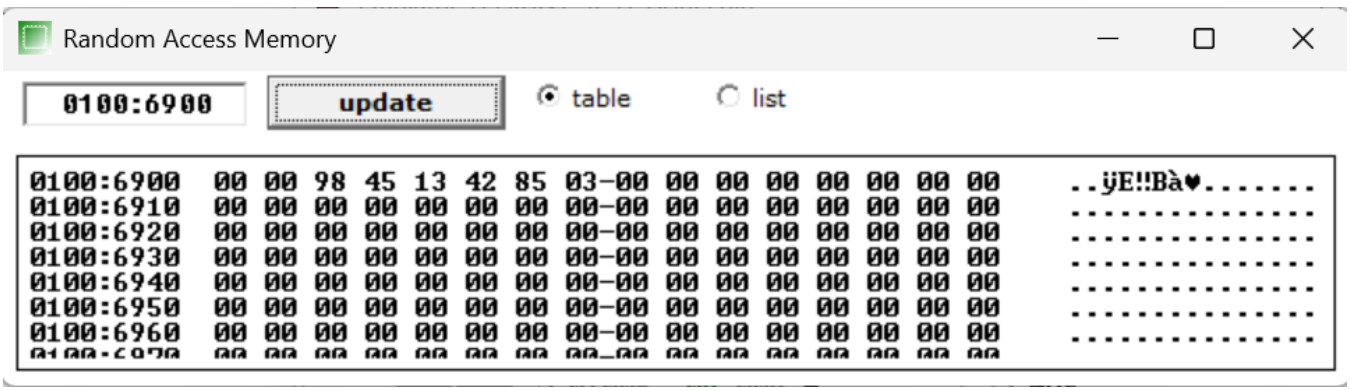
MOV CX, [6904H]

SUB AX,CX

MOV [6906H], AX

HLT



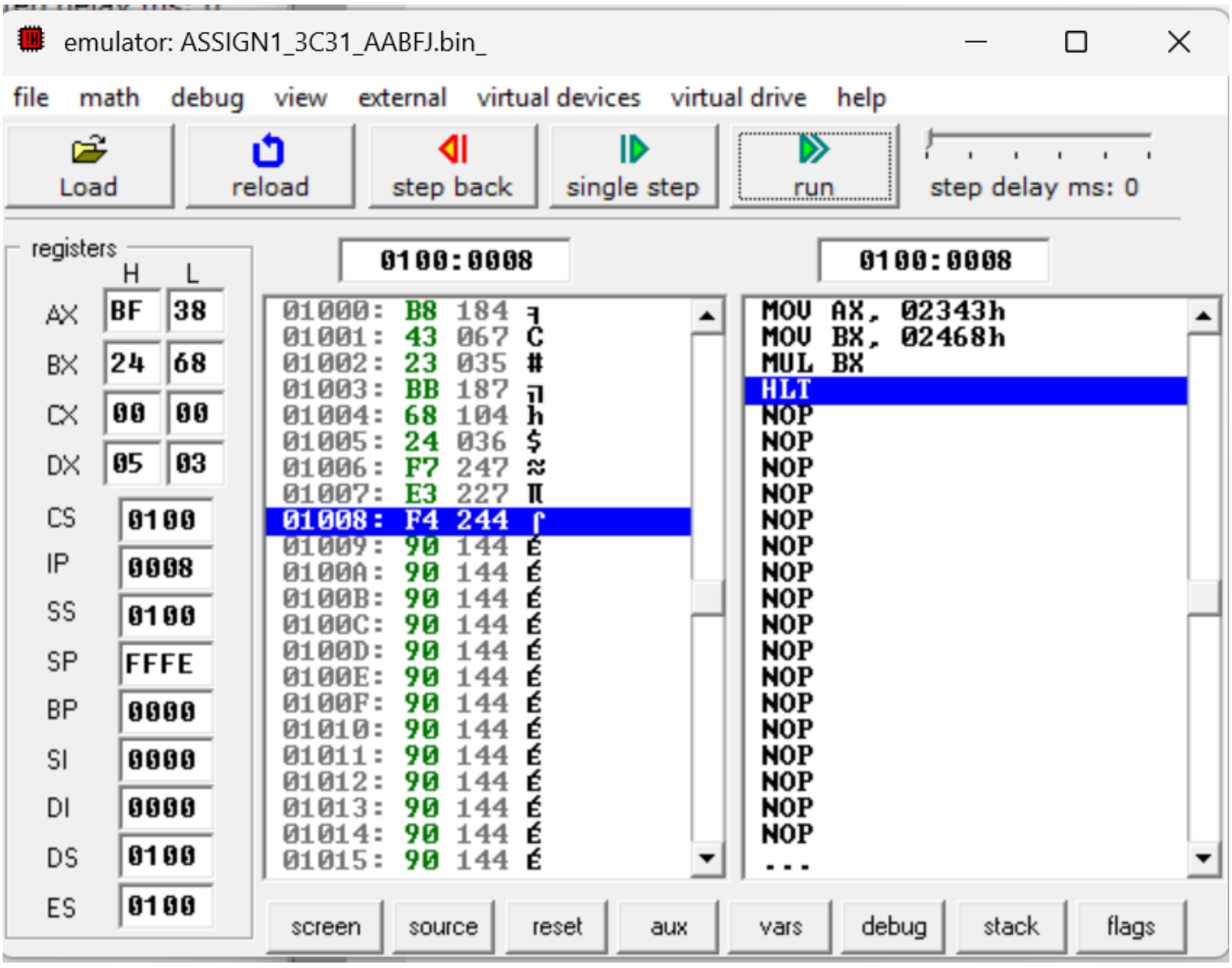


Q3.

a) Write an assembly language program to multiply two 16-bit numbers in 8086 (fixed values)

Code:

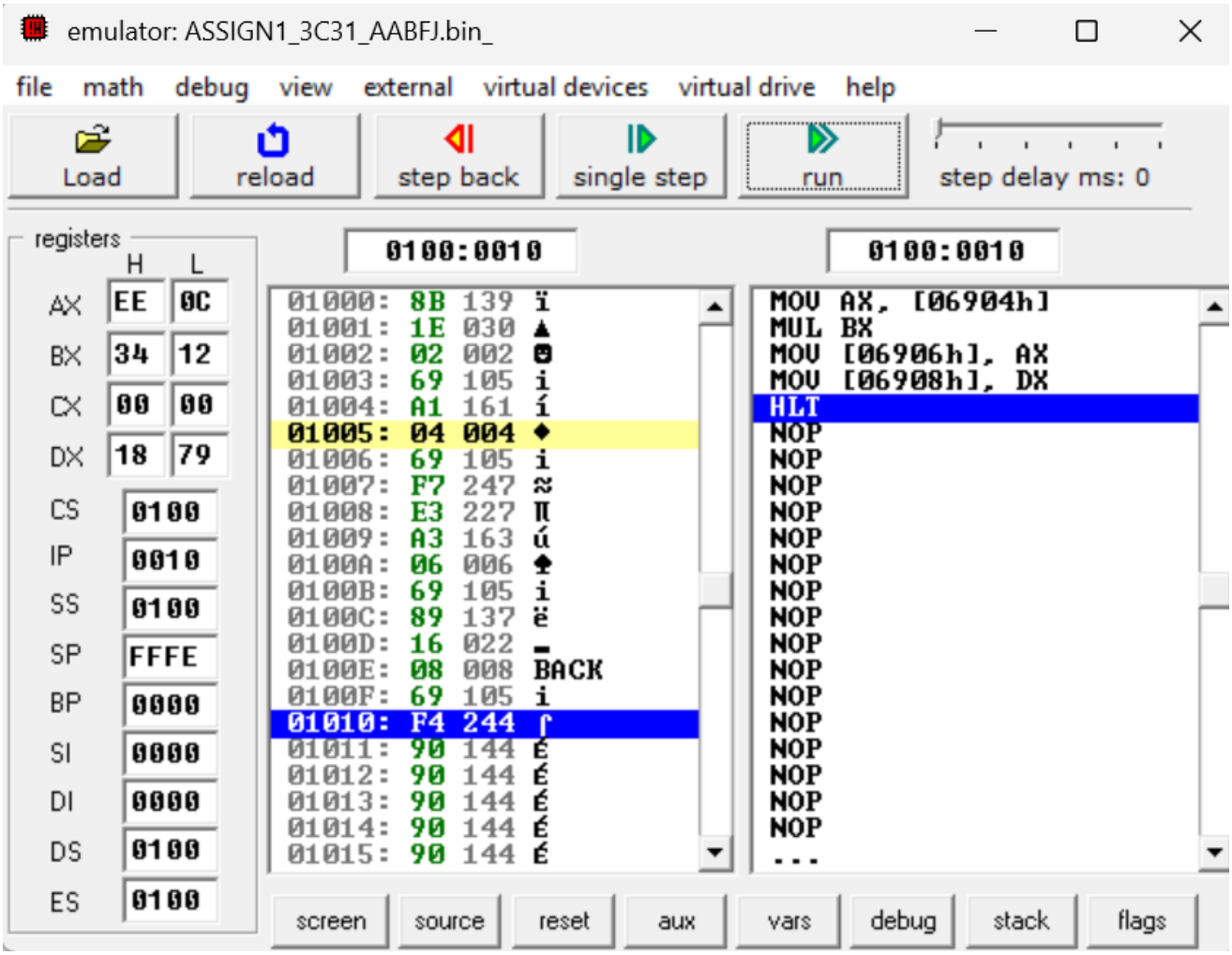
```
MOV AX, 2343H
MOV BX, 2468H
MUL BX
HLT
```

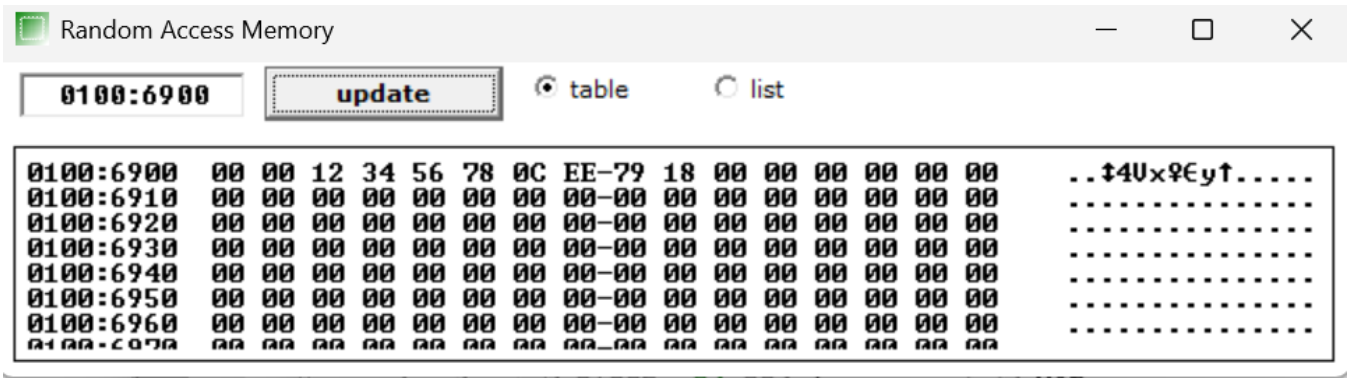


b) Multiply two 16 - bits hexadecimal numbers (runtime values)

Code:

```
MOV BX,[6902H]
MOV AX,[6904H]
MUL BX
MOV  [6906H],AX
MOV  [6908H],DX
HLT
```



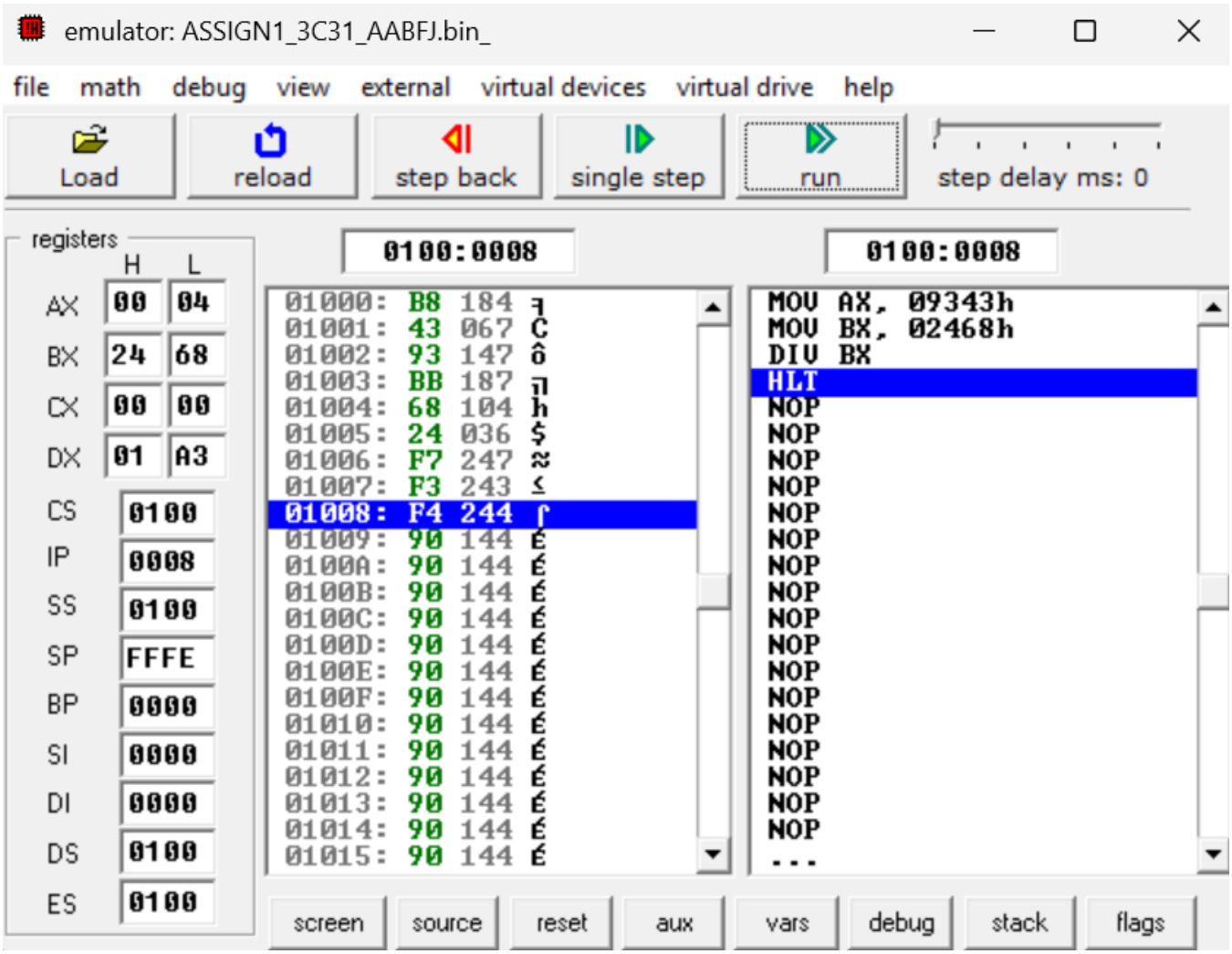


Q4.

a) Write an assembly language program to divide two 16-bit numbers in 8086. (fixed values)

Code:

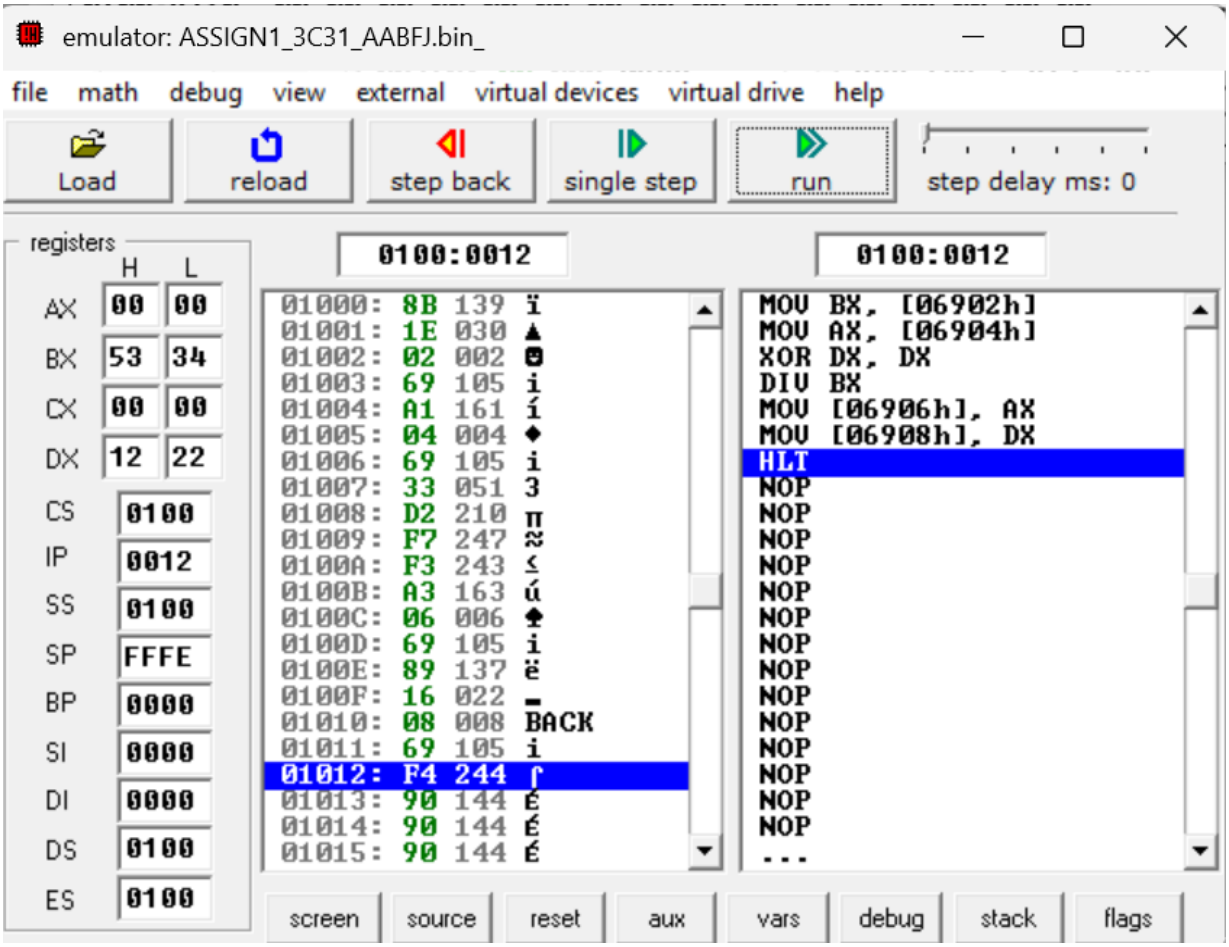
```
MOV AX, 9343H
MOV BX, 2468H
DIV BX
HLT
```

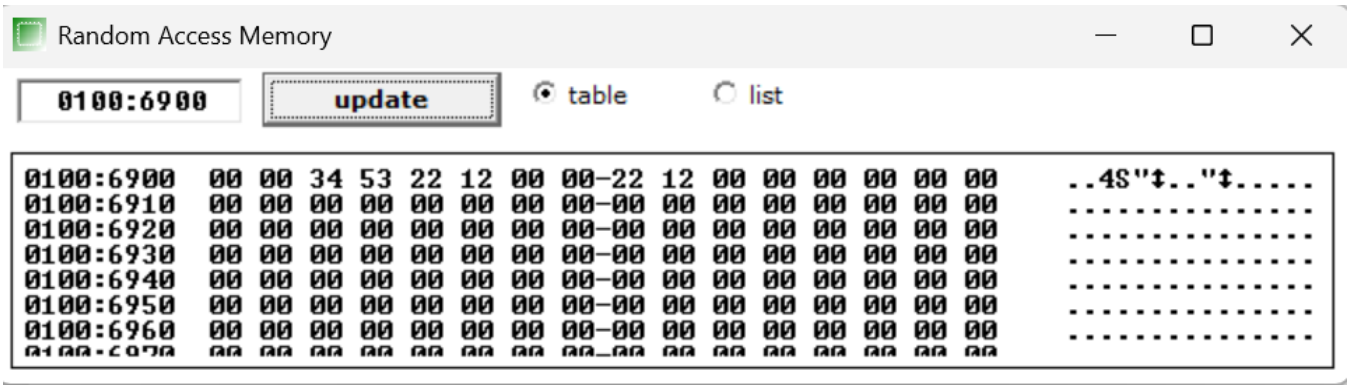


b) Divide two 16 - bits hexadecimal numbers (runtime values)

Code:

```
MOV BX,[6902H]
MOV AX,[6904H]
DIV BX
MOV  [6906H],AX
MOV  [6908H],DX
HLT
```





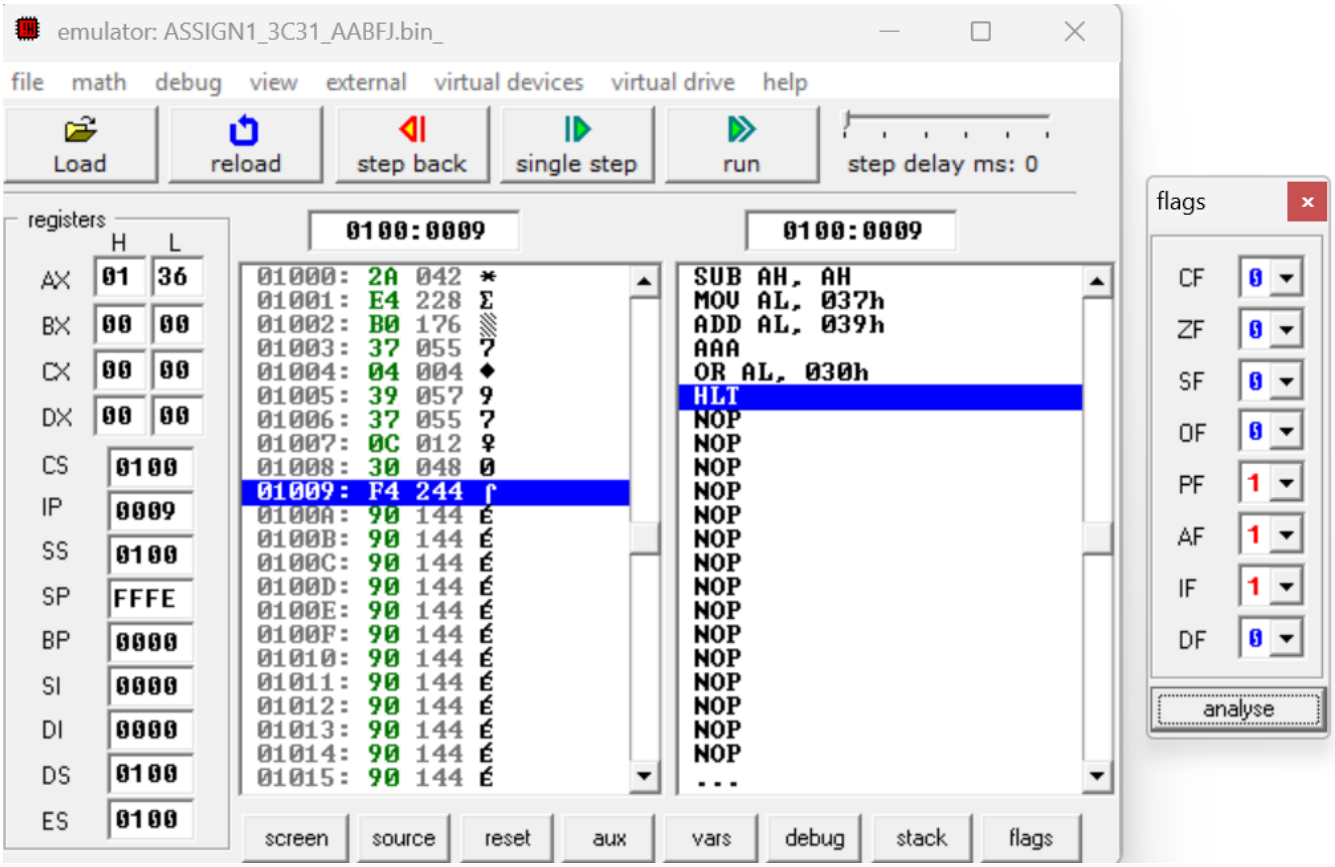
Q5.

Write an assembly language program to demonstrate AAA, AAS, AAM, AAD, DAA, and DAS in 8086.

a)AAA:

Code:

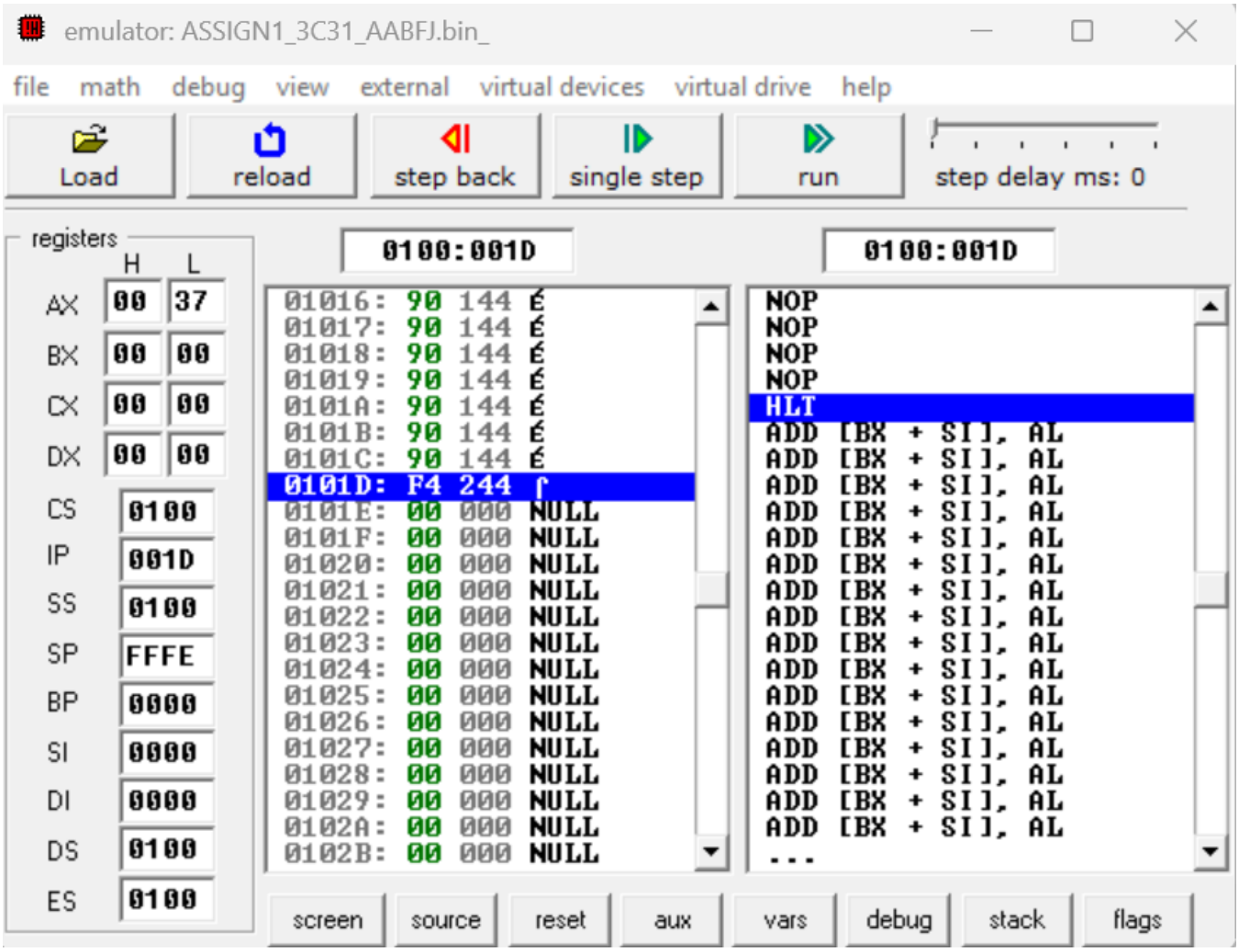
```
SUB AH,AH
MOV AL,'7'
ADD AL,'9'
AAA
OR AL,30H
HLT
```



b)AAS:

Code:

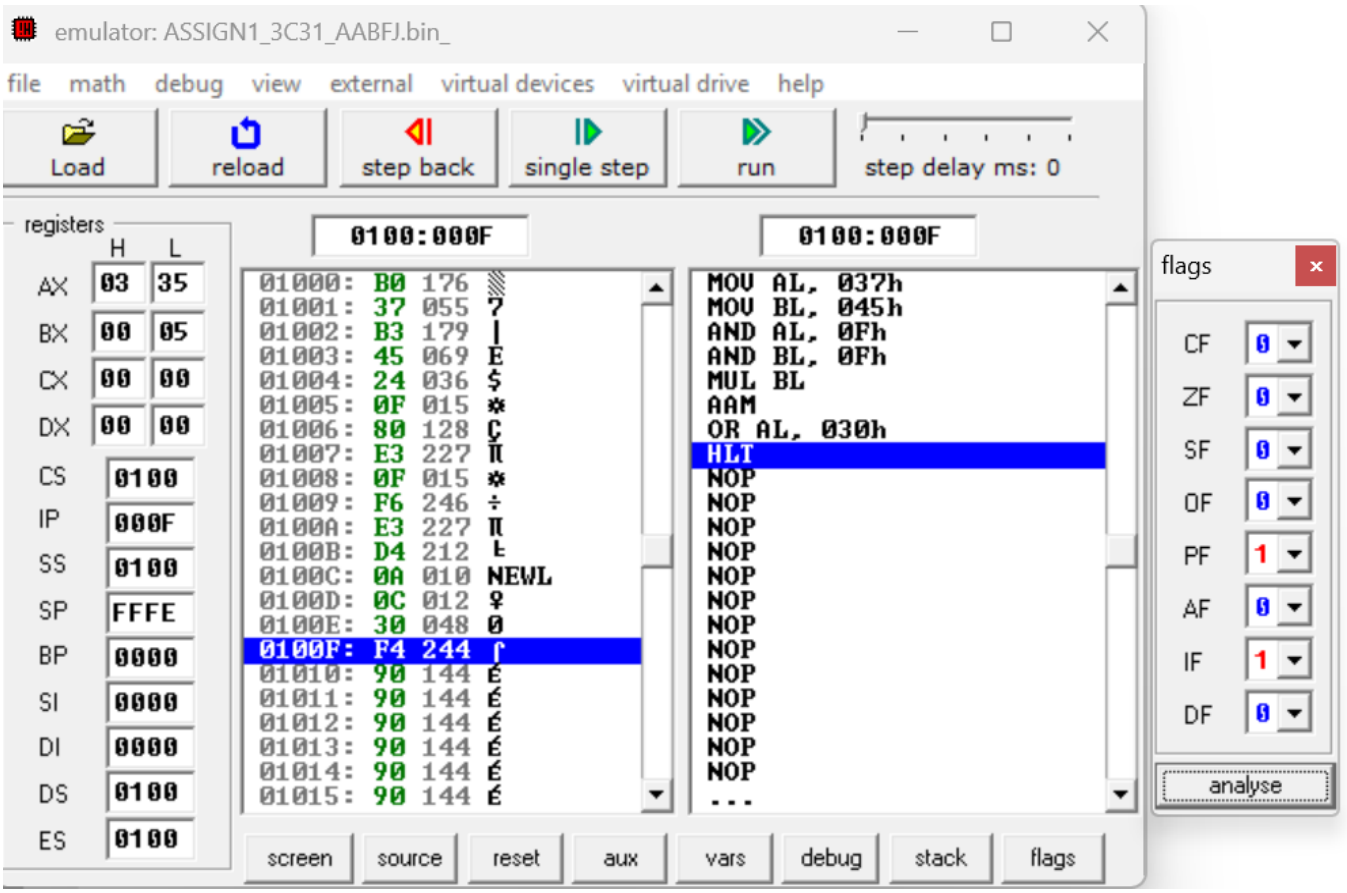
```
SUB AH,AH
MOV AL,'2'
ADD AL,'5'
AAS
OR AL,30H
```



c)AAM:

Code:

```
MOV AL,'7'
MOV BL,'E'
AND AL,0FH
AND BL,0FH
MUL BL
AAM
OR AL, 30H
HLT
```



d)AAD:

Code:

MOV AX, '74'

MOV BH, '2'

SUB AX, 3030H

SUB BH, 30H

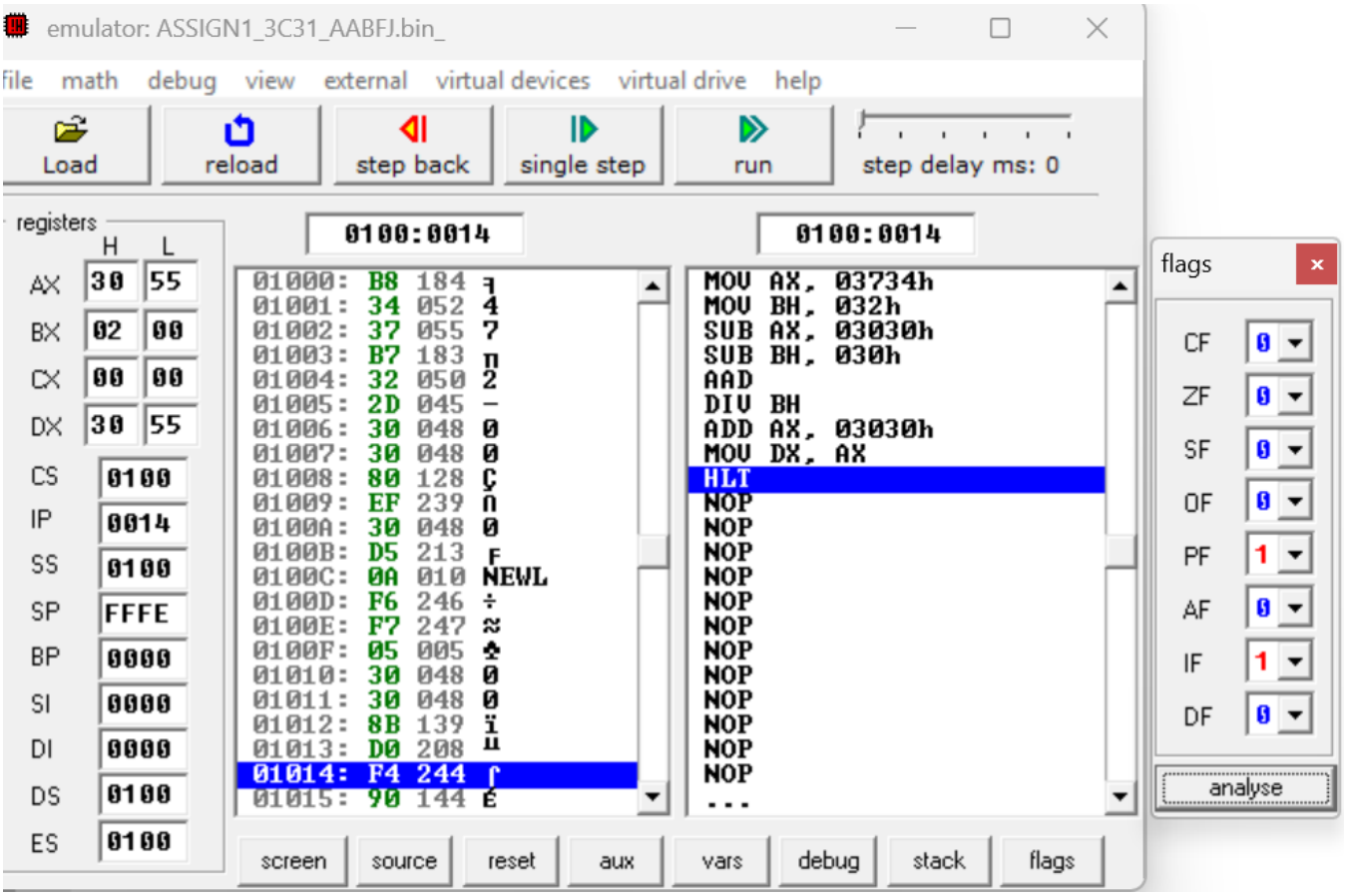
AAD

DIV BH

ADD AX, 3030H

MOV DX, AX

HLT



e)DAA:

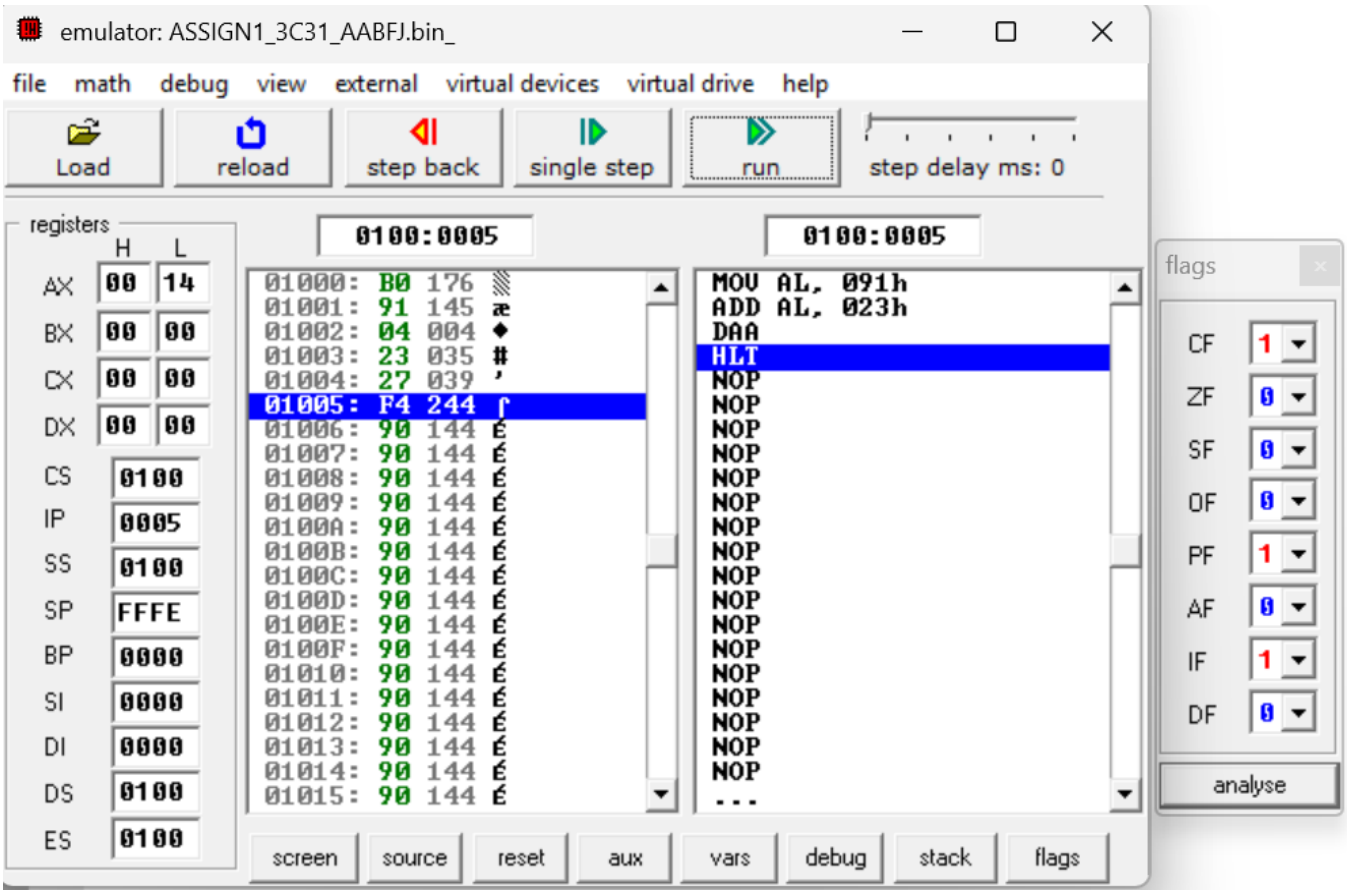
Code:

MOV AL,91H

ADD AL,23H

DAA

HLT



f)DAS:

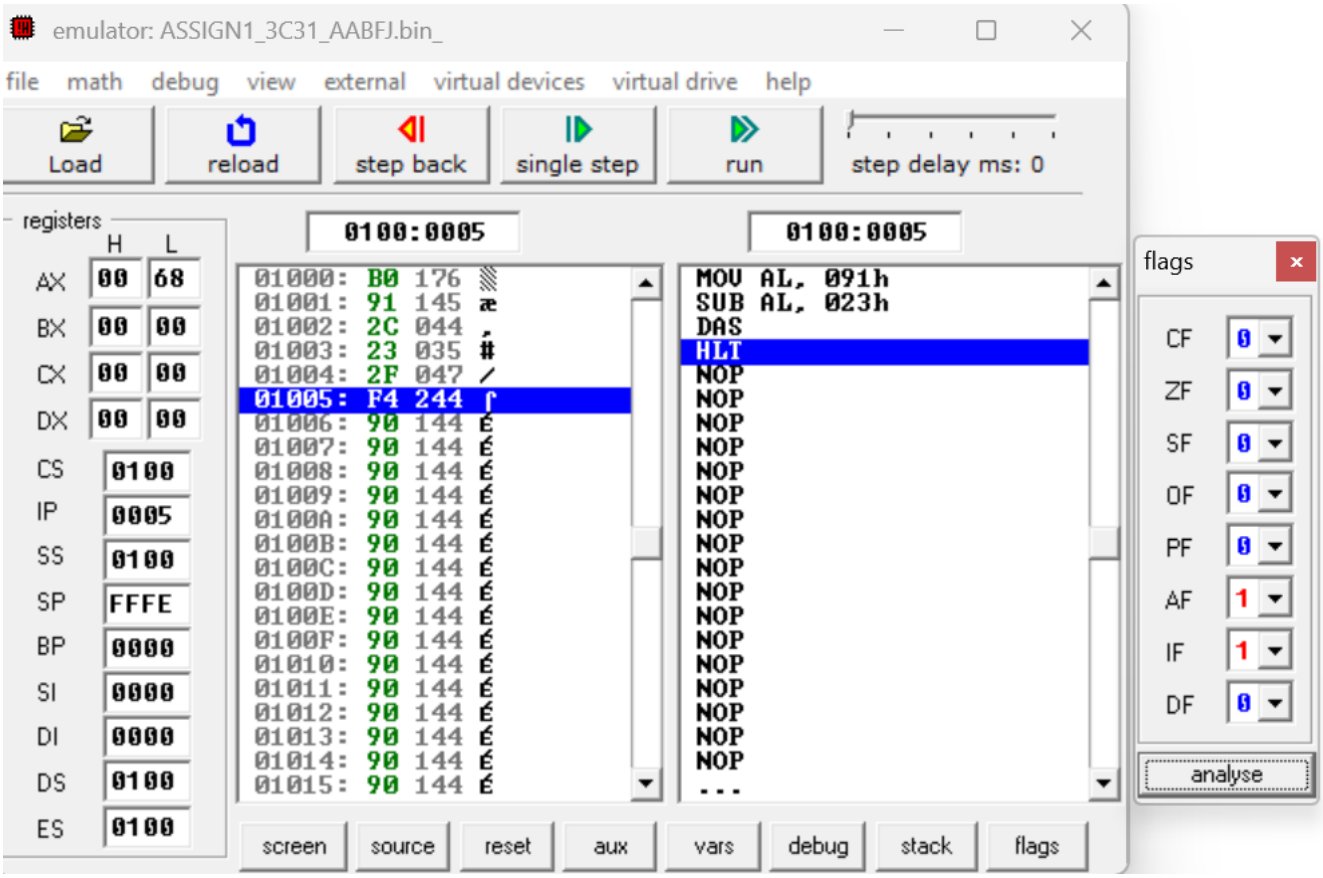
Code:

MOV AL,91H

SUB AL,23H

DAS

HLT

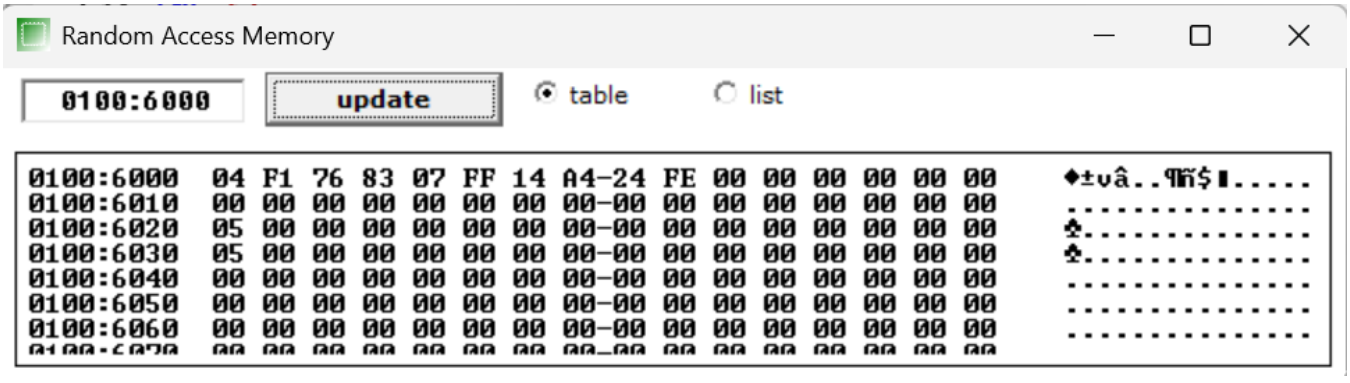


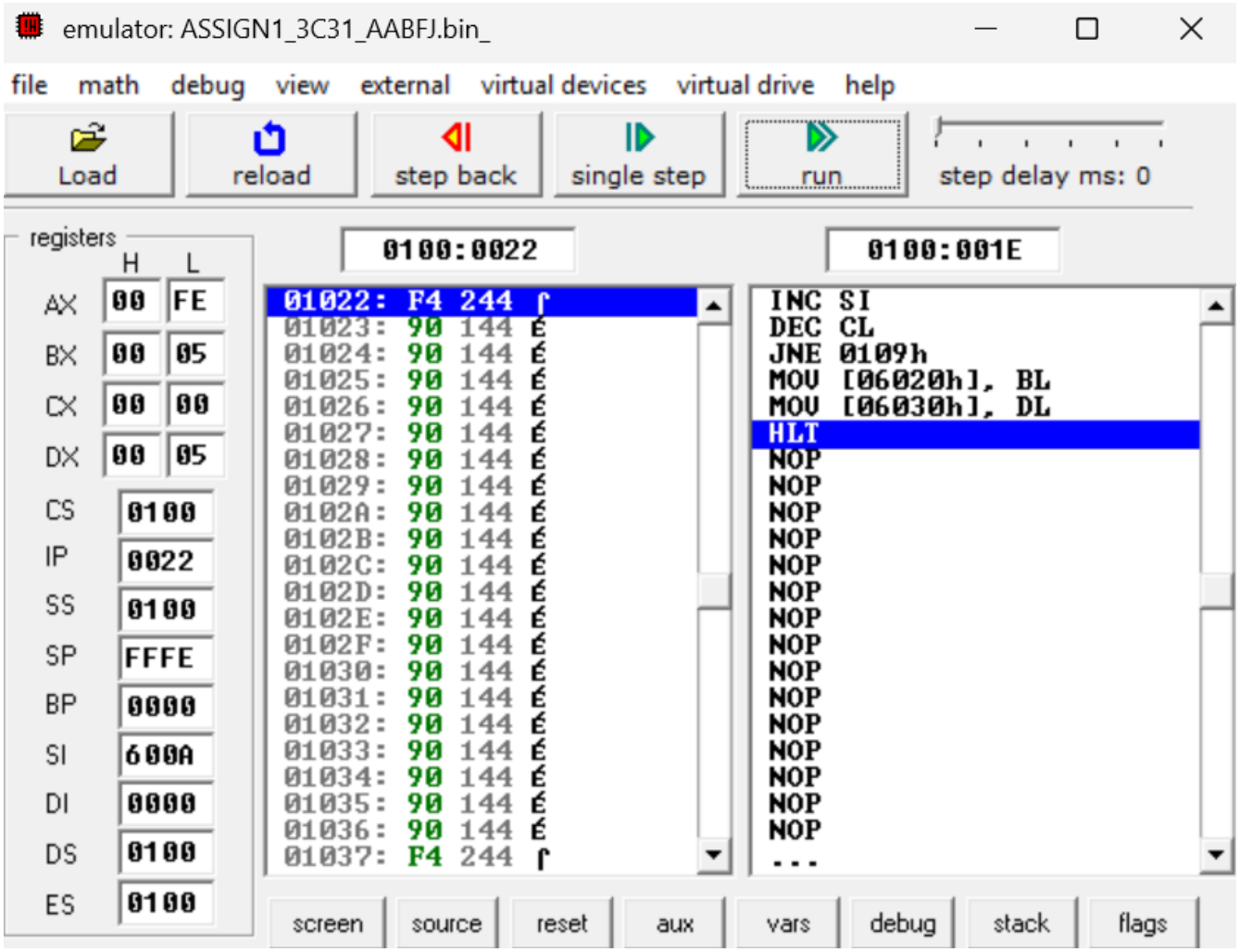
Q6.

Write an assembly language program to find out the count of positive numbers and negative numbers from a series of signed numbers in 8086.

Code:

```
MOV CL, 0AH
MOV BL, 00H
MOV DL, 00H
LEA SI,[6000H]
L1: MOV AL,[SI]
SHL AL, 01H
JNC L2
INC DL
JMP L3
L2: INC BL
L3: INC SI
DEC CL
JNZ L1
MOV [60AAH], BL
MOV [60ABH], DL
HLT
```




emulator: ASSIGN1_3C31_AABFJ.bin_
file
math
debug
view
external
virtual devices
virtual drive
help
Load
reload
step back
single step
run
step delay ms: 0
registers
H
L
AX 00 FE
BX 00 05
CX 00 00
DX 00 05
CS 0100
IP 0022
SS 0100
SP FFFE
BP 0000
SI 600A
DI 0000
DS 0100
ES 0100
0100:0022
0100:001E
01022: F4 244 ƒ
01023: 90 144 É
01024: 90 144 É
01025: 90 144 É
01026: 90 144 É
01027: 90 144 É
01028: 90 144 É
01029: 90 144 É
0102A: 90 144 É
0102B: 90 144 É
0102C: 90 144 É
0102D: 90 144 É
0102E: 90 144 É
0102F: 90 144 É
01030: 90 144 É
01031: 90 144 É
01032: 90 144 É
01033: 90 144 É
01034: 90 144 É
01035: 90 144 É
01036: 90 144 É
01037: F4 244 ƒ
INC SI
DEC CL
JNE 0109h
MOV [06020h], BL
MOV [06030h], DL
HLT
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
...
screen
source
reset
aux
vars
debug
stack
flags

Q7.

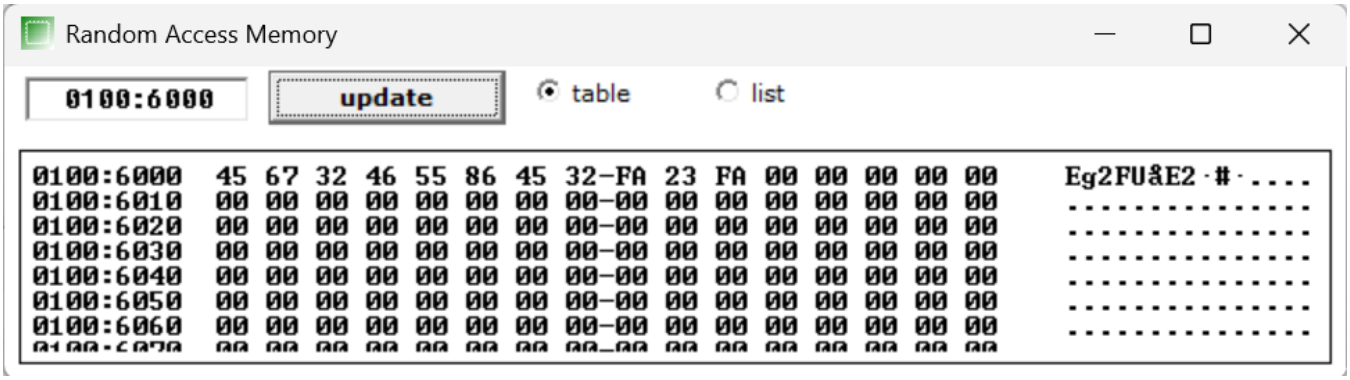
Write an assembly language program to convert to find out the largest number from a given unordered array of 8-bit numbers, stored in the locations starting from a known address in 8086.

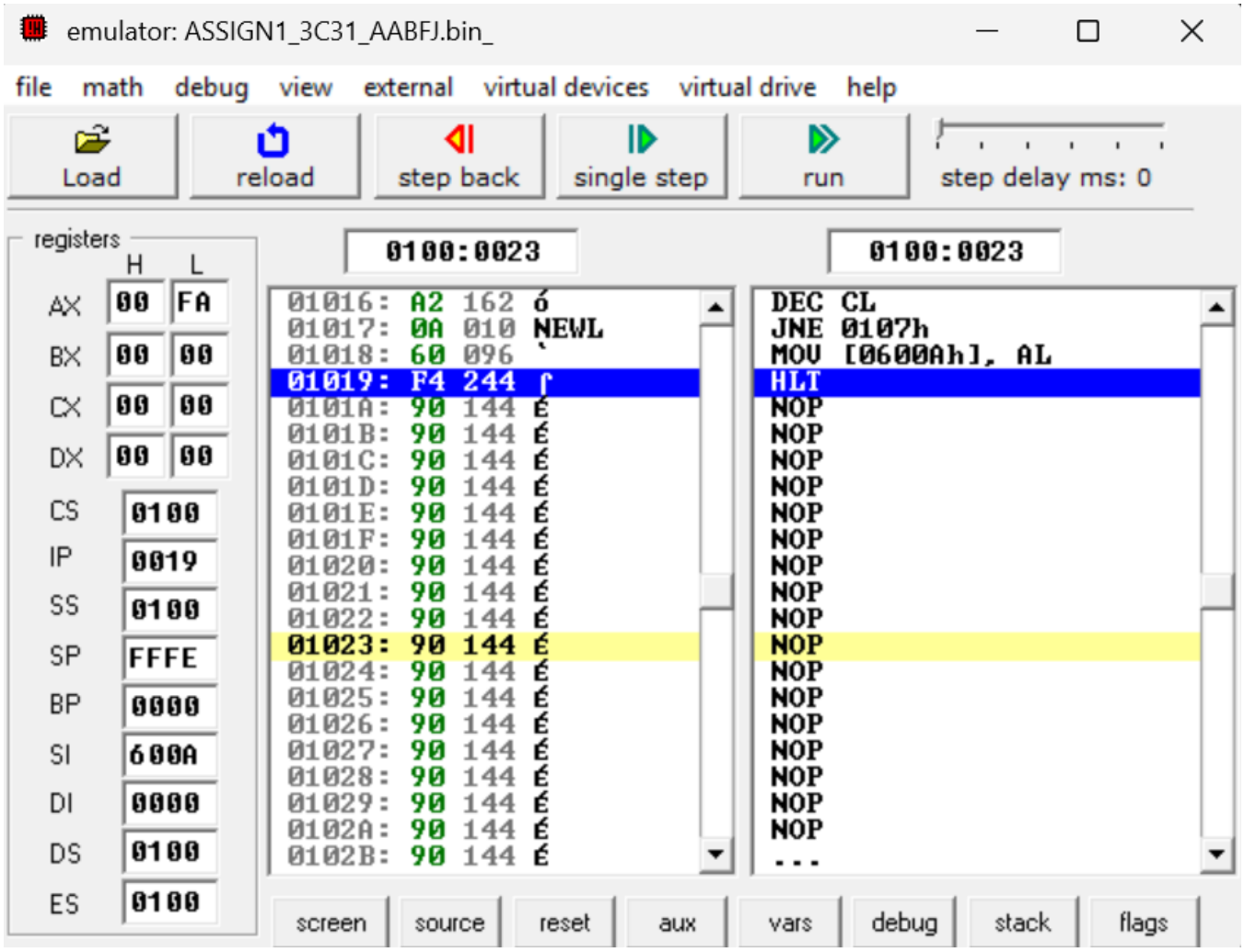
Code:

```
MOV CL,0AH

LEA SI,[6000H]
MOV AL,[SI]
L1: INC SI
MOV BL,[SI]
CMP AL,BL
JC L2
JMP L3

L2: MOV AL,BL
L3: DEC CL
JNZ L1
MOV [600AH],AL
HLT
```




emulator: ASSIGN1_3C31_AABFJ.bin_
file
math
debug
view
external
virtual devices
virtual drive
help
Load
reload
step back
single step
run
step delay ms: 0
registers
H
L
AX 00 FA
BX 00 00
CX 00 00
DX 00 00
CS 0100
IP 0019
SS 0100
SP FFFE
BP 0000
SI 600A
DI 0000
DS 0100
ES 0100
0100:0023
0100:0023
01016: A2 162 ó
01017: 0A 010 NEWL
01018: 60 096 `
01019: F4 244 ⌠
0101A: 90 144 É
0101B: 90 144 É
0101C: 90 144 É
0101D: 90 144 É
0101E: 90 144 É
0101F: 90 144 É
01020: 90 144 É
01021: 90 144 É
01022: 90 144 É
01023: 90 144 É
01024: 90 144 É
01025: 90 144 É
01026: 90 144 É
01027: 90 144 É
01028: 90 144 É
01029: 90 144 É
0102A: 90 144 É
0102B: 90 144 É
DEC CL
JNE 0107h
MOV [0600Ah], AL
HLT
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
...
screen
source
reset
aux
vars
debug
stack
flags

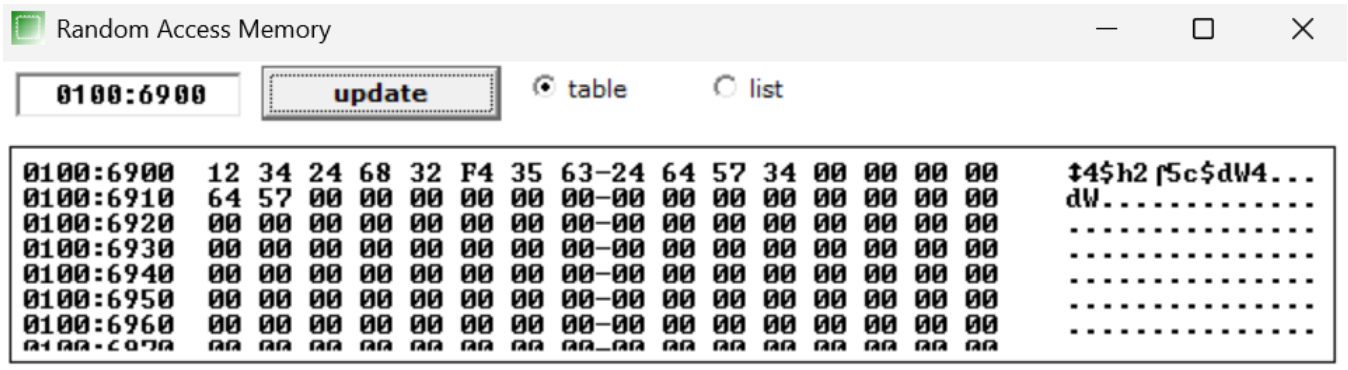
Q8.

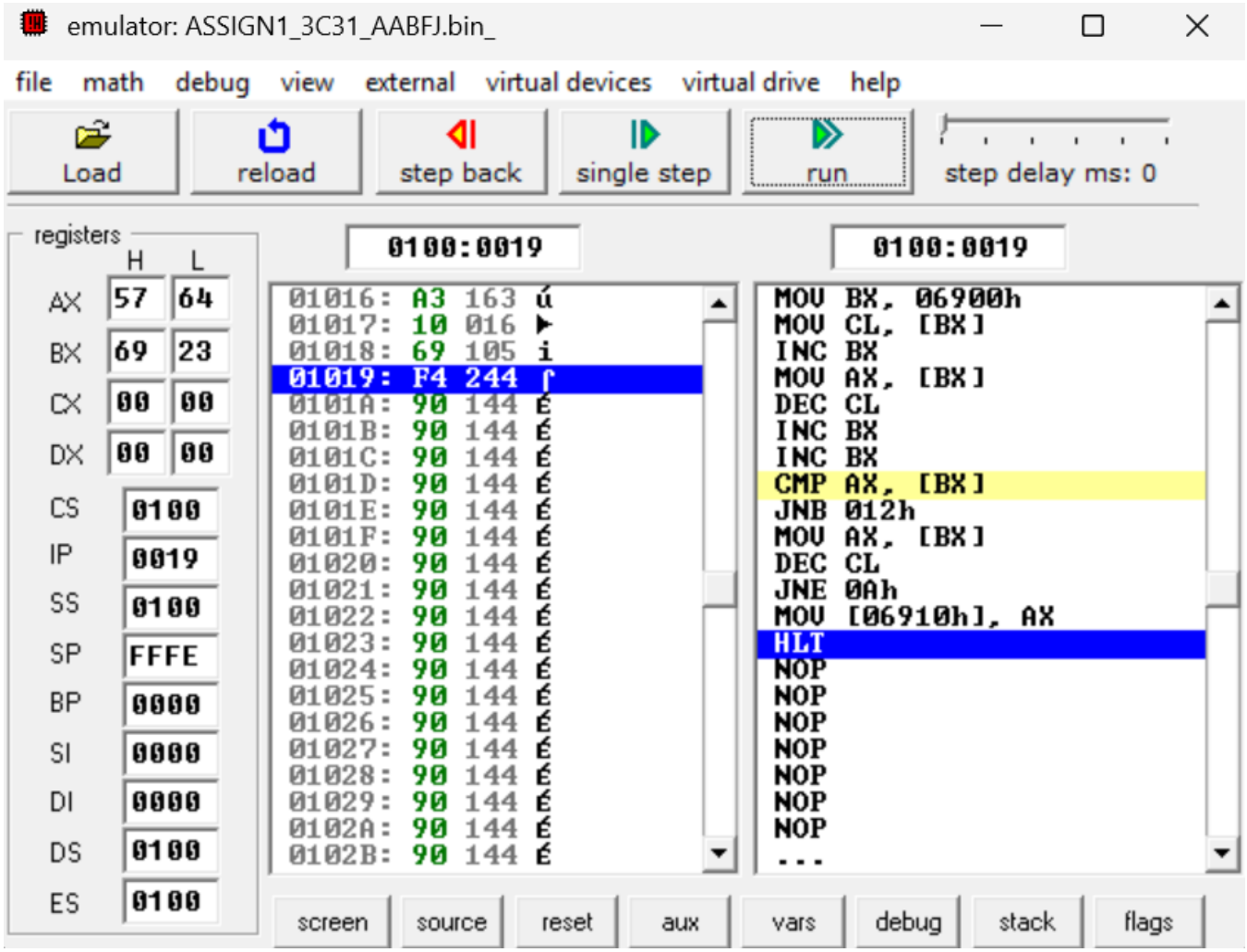
Write an assembly language program to convert to find out the largest number from a given unordered array of 16-bit numbers, stored in the locations starting from a known address in 8086.

Code:

```
MOV BX,6900H
MOV CL,[BX]
INC BX

MOV AX, [BX]
DEC CL
BACK: INC BX
INC BX
CMP AX,[BX]
JNC NEXT
MOV AX,[BX]
NEXT: DEC CL
JNZ BACK
MOV [6910H],AX
HLT
```



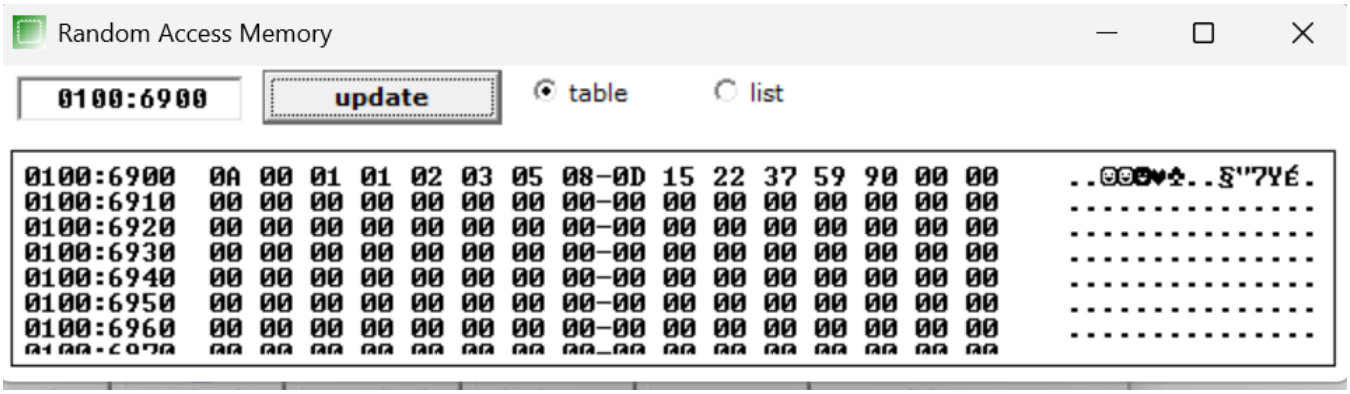

emulator: ASSIGN1_3C31_AABFJ.bin_
file
math
debug
view
external
virtual devices
virtual drive
help
Load
reload
step back
single step
run
step delay ms: 0
registers
H
L
AX 57 64
BX 69 23
CX 00 00
DX 00 00
CS 0100
IP 0019
SS 0100
SP FFFE
BP 0000
SI 0000
DI 0000
DS 0100
ES 0100
0100:0019
0100:0019
01016: A3 163 ú
01017: 10 016 ►
01018: 69 105 i
01019: F4 244 ⌠
0101A: 90 144 É
0101B: 90 144 É
0101C: 90 144 É
0101D: 90 144 É
0101E: 90 144 É
0101F: 90 144 É
01020: 90 144 É
01021: 90 144 É
01022: 90 144 É
01023: 90 144 É
01024: 90 144 É
01025: 90 144 É
01026: 90 144 É
01027: 90 144 É
01028: 90 144 É
01029: 90 144 É
0102A: 90 144 É
0102B: 90 144 É
MOV BX, 06900h
MOV CL, [BX]
INC BX
MOV AX, [BX]
DEC CL
INC BX
INC BX
CMP AX, [BX]
JNB 012h
MOV AX, [BX]
DEC CL
JNE 0Ah
MOV [06910h], AX
HLT
NOP
NOP
NOP
NOP
NOP
NOP
NOP
...
screen
source
reset
aux
vars
debug
stack
flags

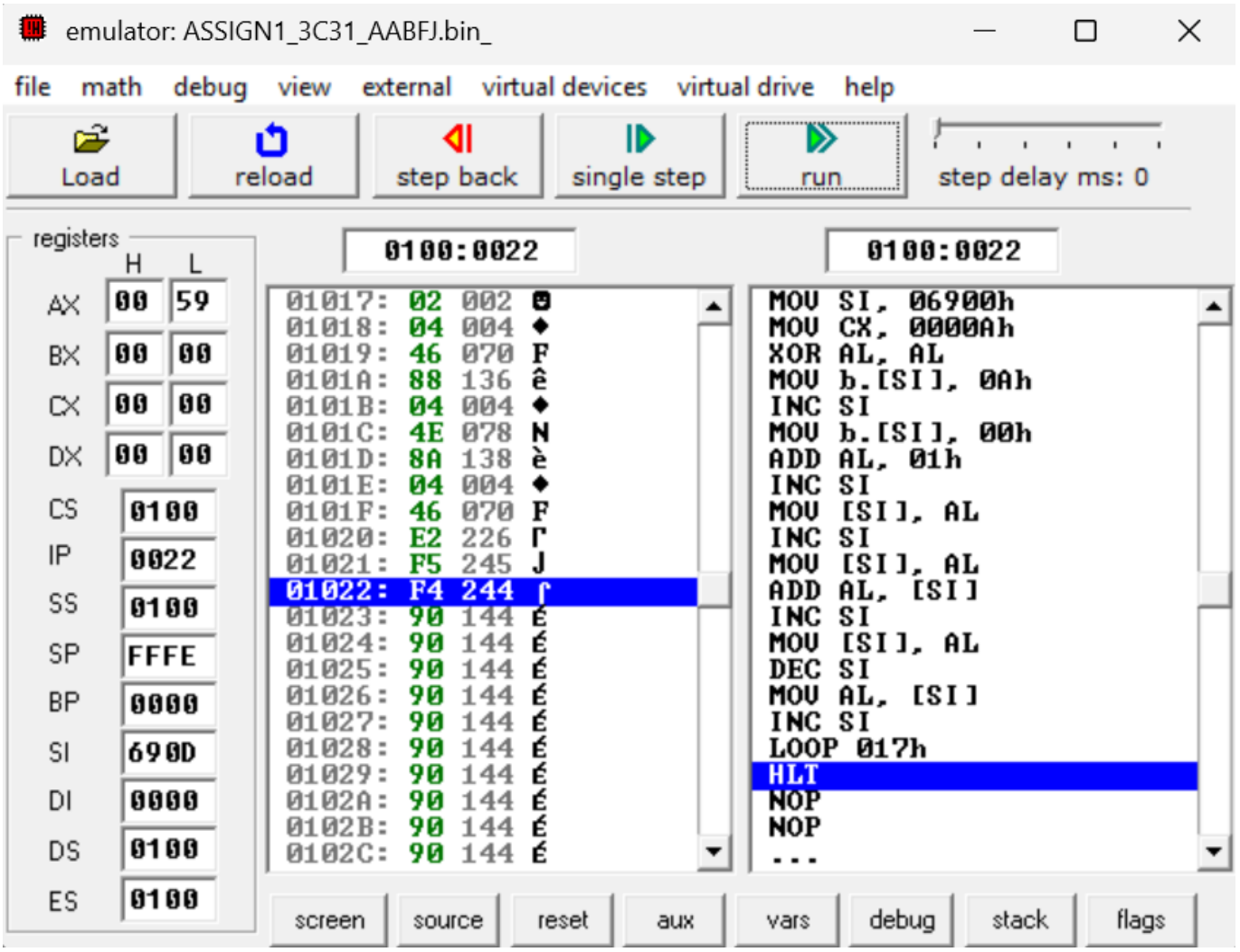
Q9.

Write an assembly language program to print Fibonacci series in 8086.

Code:

```
MOV SI, 6900H
MOV CX, 0AH
XOR AL, AL
MOV [SI], 0AH
INC SI
MOV [SI], 00H
ADD AL, 01H
INC SI
MOV [SI], AL
INC SI
MOV [SI], AL
BACK: ADD AL,[SI]
INC SI
MOV [SI], AL
DEC SI
MOV AL, [SI]
INC SI
LOOP BACK
HLT
```




emulator: ASSIGN1_3C31_AABFJ.bin_
file
math
debug
view
external
virtual devices
virtual drive
help
Load
reload
step back
single step
run
step delay ms: 0
registers
H
L
AX 00 59
BX 00 00
CX 00 00
DX 00 00
CS 0100
IP 0022
SS 0100
SP FFFE
BP 0000
SI 690D
DI 0000
DS 0100
ES 0100
0100:0022
01017: 02 002
01018: 04 004
01019: 46 070 F
0101A: 88 136
0101B: 04 004
0101C: 4E 078 N
0101D: 8A 138
0101E: 04 004
0101F: 46 070 F
01020: E2 226
01021: F5 245 J
01022: F4 244
01023: 90 144 É
01024: 90 144 É
01025: 90 144 É
01026: 90 144 É
01027: 90 144 É
01028: 90 144 É
01029: 90 144 É
0102A: 90 144 É
0102B: 90 144 É
0102C: 90 144 É
0100:0022
MOV SI, 06900h
MOV CX, 0000Ah
XOR AL, AL
MOV b.[SI], 0Ah
INC SI
MOV b.[SI], 00h
ADD AL, 01h
INC SI
MOV [SI], AL
INC SI
MOV [SI], AL
ADD AL, [SI]
INC SI
MOV [SI], AL
DEC SI
MOV AL, [SI]
INC SI
LOOP 017h
HLT
NOP
NOP
...
screen
source
reset
aux
vars
debug
stack
flags

Q10.

Write an assembly language program to perform the division 15/6 using the ASCII codes. Store the ASCII codes of the result in register DX.

Code:

```
MOV AH,'1'
MOV AL,'5'
MOV BH,'6'
SUB AX,3030h
SUB BH,30H
AAD
DIV BH
ADD AX,3030H
MOV DX,AX
HLT
```